HOORVIA - OHM'S LAW & POWER FORMULAS CHART: PRINTABLE ELECTRICAL EQUATIONS GUIDE

Quick Reference Cheat Sheet · US Letter Printable

Crafting & DIY · Sheet #039

Formula / What It Finds	Standard Equation	Solve-For Variants	SI Units	Worked Example
Ohm's Law (Voltage Drop)	V = I x R	I = V / R ; R = V / I	Volts (V)	A 10.0 ? heating element drawing 12.0 A needs V = 12.0 x 10.0 = 120.0 V
Ohm's Law (Current Draw)	I = V / R	V = I x R ; R = V / I	Amperes (A)	120 V across 15.0 ? draws I = 120 / 15.0 = 8.0 A
Ohm's Law (Resistance)	R = V / I	V = I x R ; I = V / R	Ohms (?)	A lamp drawing 0.50 A on 120 V has R = 120 / 0.50 = 240.0 ?
Watt's Law (Power)	P = V x I	I = P / V ; V = P / I	Watts (W)	120 V x 12.0 A = 1,440.0 W (80% of a 15 A circuit)
Joule's Law (I²R Heating)	P = I² x R	I = ?(P / R) ; R = P / I²	Watts (W)	15.0 A through 0.40 ? wastes 15² x 0.40 = 90.0 W as heat
Voltage-Resistance Power	P = V² / R	V = ?(P x R) ; R = V² / P	Watts (W)	120 V across 8.0 ? gives (120)² / 8.0 = 1,800.0 W
Solve for Voltage (Power & Current)	V = P / I	P = V x I ; I = P / V	Volts (V)	A 1,500 W heater drawing 12.5 A runs at V = 1,500 / 12.5 = 120.0 V
Solve for Current (Power & Resistance)	I = ?(P / R)	P = I² x R ; R = P / I²	Amperes (A)	A 900 W element of 10.0 ? draws I = ?(900 / 10) = 9.49 A
Solve for Resistance (Power & Current)	R = P / I²	I = ?(P / R) ; P = I² x R	Ohms (?)	A 1,200 W load at 10.0 A has R = 1,200 / 100 = 12.0 ?
Solve for Resistance (Voltage & Power)	R = V² / P	V = ?(P x R) ; P = V² / R	Ohms (?)	A 60 W bulb on 120 V has R = 14,400 / 60 = 240.0 ?
Single-Phase AC Real Power	P = V x I x PF	PF = cos(?) = P / (V x I) ; S = V x I (VA)	Watts (W) vs Volt-Amps (VA)	A 120 V motor at 10.0 A, 0.85 PF delivers 120 x 10.0 x 0.85 = 1,020.0 W
Three-Phase AC Real Power	P = ?3 x V_LL x I_L x PF	?3 ? 1.732 ; I_L = P / (?3 x V_LL x PF)	Watts (W) / Kilowatts (kW)	A 480 V feeder at 30.0 A, 0.90 PF delivers 1.732 x 480 x 30.0 x 0.90 = 22,447 W (22.45 kW)

PRO TIP: Memorize the triangle: Cover the letter you want in the V-I-R triangle; the remaining two reveal the formula - the fastest recall.